

Budapest University of Technology and Economics

CoARA Action Plan

Organization	Budapest University of Technology and Economics
Responsible person	Rector
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About BME

The Budapest University of Technology and Economics (BME) is one of Hungary's leading higher education institutions, with a history covering more than 240 years. BME offers a wide range of courses in engineering, information technology, economics, and natural sciences and is committed to high-quality teaching and research. BME aims to equip its students with state-of-the-art knowledge and skills to succeed in the global labour market.

BME's international reputation is based on its excellent teaching and research staff and an exceptional involvement in national and international research projects. BME's research activities range from basic research and developing cutting-edge technologies to industrial cooperation and innovation. The university maintains close links with national and international industrial partners and ensures that its research output entails direct industrial applications.

With its 21,000 students and 1,200 full-time lecturers and researchers, BME plays a central role in Hungarian higher education and research in engineering and actively contributes to scientific and technological development in Central Europe. BME has eight faculties hosting a diverse educational program and a wide range of research laboratories. The university's student and faculty community is dynamic and diverse and provides an inspiring environment for study and research. BME is committed to continue to be a leader in engineering and business education and to contribute to Europe's sustainable future.

Synopsis of current evaluation practices

BME has embraced the principles of Responsible Research Assessment¹ and is committed to its promotion both nationally and internationally. The University is also committed to integrating fair and responsible assessment in its internal evaluation processes. At the same time, BME recognizes the corresponding initiatives of the Hungarian Accreditation Board and the Hungarian Academy of Sciences, as well as CoARA.

Our current practices for measuring research performance are as follows:

1. Only bodies endowed with appropriate knowledge (the members of which have at least a PhD degree) may assess research performance.
2. In order to reduce bias and partiality, the participation of external members – not employed by the BME and possibly of foreign nationality – should also be included in the appropriate panels. In all cases, the composition of the bodies set up under the BME Doctoral and Habilitation Procedures shall comply with these principles.
3. BME's University Habilitation Committee and Doctoral Council (EHBDT), the specialty Disciplinary Habilitation Committee and Doctoral Council (HBDT), and the Council of Doctoral Schools (DITs) oversee research performance evaluation in the case of conferring PhD degrees, habilitations, the honorary doctorates and the Neumann Professorships. These committees are also responsible for evaluating research performance prior to the appointment of university professors, scientific advisors, and research professors.
4. When evaluating research performance, *peer review/evaluation plays a central role*. Quantitative measures are mainly used to assess whether the candidate meets certain *minimal criteria*. Minimal criteria are established such that they can be accomplished in all career paths in every professional field. Certain career paths and research areas do not allow the completion of these minimal criteria. In these cases the panels can disregard them based on merits and on the conceptual and well-documented considerations. Meeting the minimal criteria is not a sufficient, but only a *necessary condition* for a positive decision.
5. Different assessment metrics should be applied at different stages of the career path and for different research areas to reflect the diversity of researchers and to ensure equal opportunities in their academic progress. At the beginning of a researcher's career, e.g., citation rates are less relevant and research quality should rather be assessed by journal ratings. At a later stage of the research career (habilitation, professorship), scientific impact such as invited talks or citations should be considered.

The primary measure of research quality should be based on a *comprehensive analysis* of the *validity*, the *scientific value*, the *novelties*, and the *impact* of the results. Moreover, *social impact* and relevance for Sustainable Development Goal(s) must also be integral parts of research quality assessment.

¹ <https://coara.eu/agreement/the-agreement-full-text/>

BME is committed to ensuring that the evaluation of research, researchers, and research organizations adopt to the diversity of practices and activities, and that help to improve the quality and impact of research.

By joining CoARA, BME expressed its conviction that methodologies for measuring and evaluating scientific performance need to be reviewed both within BME and the wider scientific community.

BME agrees with the following commitments proposed in the Agreement, and is devoted to update its internal evaluation processes.

CoARA REFLECTION POINTS	BME ACTIONS
Reflect on your strategy and change approach	Involve BME's Scientific Council to study existing BME practices.
Involve your institutional community in the change process	BME's Scientific Council (SC) is a body that represents BME's wide scientific community. The SC will coordinate the CoARA action plan, issue recommendations, identify strategical directions, and publish these on BME's web page, and present these towards BME's top management.
Identify key challenges to address	BME's SC will overview existing practices and specify recommended changes, especially in the context of Gender Equality, early career research assessment, and the internationalization of the assessment process.
Recognize the diversity of contributions to, and careers in, research in accordance with the needs and nature of the research	At BME all disciplines have equal opportunities in terms of accessing research funding. Access to facilities and tools should be guaranteed for all members of BME's research staff. BME consistently promotes equal opportunities without discrimination based on gender, career stage or research area. BME is currently in the process of transforming its structure and operation, as well as its legal status. In course of this transformation, special emphasis shall be placed on launching internal programs to support diverse career paths and ensuring

	that younger and under-represented groups of researchers have fair development opportunities in the academic community. At BME, research contributions shall not be limited to traditional publications, and personal evaluations will equally include components such as recognizing all forms of research activities as well as public engagement and technology transfer.
Base research assessment primarily on qualitative evaluation of which peer review is central, supported by responsible use of quantitative indicators	BME's current practices in promotion as well as job assignments are currently based on peer review and a complex assessment. We shall enforce and sustain these good practices.
Abandon inappropriate uses in research assessment of journal- and publication-based metrics, in particular inappropriate uses of Journal Impact Factor (JIF) and h-index	Currently, BME does not apply inappropriate journal- and publication-based metrics in research assessment. However, with the advent of international rankings, we experience the danger of a swift transition towards a purely metrics-based research assessment. BME will therefore strengthen non-metrics-based, but rather peer-review related components in its evaluation processes, raise awareness against this danger in its entire research community. Recruiting principles shall be revised focusing on comprehensive research quality, instead of considering only quantitative factors, encouraging the consideration of a wide range of research results. BME shall also test alternative assessment formats such as narrative CV, competency-based CV, and evidence-based CV formats.
Avoid the use of rankings of research organisations in research assessment	BME plans to open towards international job markets in coming years and establish a purely merit-based evaluation procedure.
Commit resources to reforming research assessment as is needed to achieve the organisational changes committed to	The BME's top management envisages a new operating model. In this new model, a new position of Vice-Rector for Science has been created, and a Scientific Program Office is

	being established. These positions/resources shall support achieving BME's CoARA goals.
Review and develop research assessment criteria, tools and processes	BME will lay down Research Assessment Principles. The RAP will provide guidance and support to assessment committees and juries at BME.
Raise awareness of research assessment, reform and provide transparent communication, guidance, and training on assessment criteria and processes as well as their use	The RAP shall be disseminated within BME's research community, and made public on BME's website.
Exchange practices and experiences to enable mutual learning within and beyond the Coalition	As an active member of CoARA's National Chapter, we shall share our best practices within Hungary as well as with our international partners.
Communicate progress made on adherence to the principles and implementation of the Commitments	BME's commitment to open communication and collecting feedback is crucial for the success of the CoARA reform. Therefore, elements contributing to the reform will be discussed and debated internally. The collected results will be presented to the academic community and feedback and suggestions will be requested. The approval of the reform will be carried out by Academic Bodies such as the University Habilitation Committee and Doctorial Council, and supported by BME's Scientific Council.
Evaluate practices, criteria and tools based on solid evidence and the state-of-the-art in research on research, and make data openly available for evidence gathering and research.	In the previous period, BME's Scientific Council created a comprehensive "Open Science Strategy". BME is committed to launch the actions highlighted in this document, and enforce its open data and open science policy in forthcoming years.

Roadmap 2023-2027

2023	BME became an official signatory of the CoARA Agreement
2024	The BME's designated representatives followed CoARA's online events and attended the general assembly. BME has prepared its CoARA Action Plan.
2025	Finalizing and launching BME's CoARA Action Plan. Continuous improvement of BME procedures. Active participation in national and international organizations, networks and working groups. Launching BME's Open Science Plan.
2026	Continuous improvement of BME procedures. Active participation in national and international organizations, networks and working groups. Assess and evaluate further development objectives.
2027	Review of the BME CoARA Action Plan.